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Quantifying the impact of different precipitation data sources on hydrological modeling processes in arid basin using transfer entropy

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ABSTRACT

Multi-source precipitation products (MSPs) are critical for hydrologic modeling, but their spatial and temporal heterogeneity and uncertainty present challenges to simulation accuracy that need to be addressed urgently. This study assessed the impact of different precipitation data sources on hydrologic modeling in an arid basin. There were seven precipitation products and meteorological station interpolated data that were used to drive the hydrological model, and we evaluated their performance by fusing the six precipitation products through the dynamic bayesian averaging algorithm. Ultimately, the runoff simulation uncertainty was quantified based on the DREAM algorithm, and the information transfer entropy was used to quantify the differences in hydrologic simulation processes driven by different precipitation data. The results showed that CMFD and ERA5 weights were higher, and the DBMA fused precipitation annual mean value was about 309.83 mm with good simulation accuracy (RMSE of 1.46 and R^2 of 0.75). The simulation was satisfactory (NSE >0.80) after parameter calibration and data assimilation for all driving data, with CHIRPS and TRMM performed better in the common mode, and HRLT and CMFD performed excellently in the glacier mode. The DREAM algorithm indicated less uncertainty for DBMA, CHIRPS and HRLT data. The entropy of information transfer revealed that precipitation occupied a significant position in information transfer, especially affecting evapotranspiration and surface soil moisture. CMFD and TPS CMADS were highest in snow water equivalent information entropy, and CHIRPS and TPS CMADS were highest in evapotranspiration information entropy. CDR, CHIRPS, ERA5-Land and IDW STATION had the highest snow water equivalent information entropy, DBMA and CMORPH had the highest runoff information entropy, CHIRPS and TRMM had the highest soil moisture information entropy, whereas ERA5, HRLT, and TPS CMADS had the highest evapotranspiration information entropy in glacial mode. This study reveals significant differences between different precipitation data sources in hydrological modeling of arid basin, which is an important reference for future water resources management and climate change adaptation strategies.

1. Introduction

Precipitation as a critical driving factor in hydrologic modeling, significantly influences surface hydrologic processes and plays an important role in water resources management (Schreiner-McGraw and Ajami, 2020). However, due to its high spatial and temporal heterogeneity, station observations are unable to satisfy the need for high-precision hydrological modeling (Han et al., 2020). With the development of remote sensing technology, an increasing number of satellite precipitation datasets have been used in recent decades as hydrologic modeling-driven data, to complement the mountainous areas that are difficult to cover with rain gauges and to provide relatively homogeneous precipitation estimates (Chen et al., 2020; Wang et al., 2020).

Rapidly evolving satellite technologies and atmospheric models provide new opportunities for precipitation estimation, and multi-source precipitation products (MSPs), such as CHIRPS, CMFD, CMORPH, ERA5, and PERSIANN-CDR, are widely used for hydrologic modeling and extreme precipitation detection due to their high spatial and temporal accuracy and ease of use (Huffman et al., 2007; Joyce et al., 2004; Lei et al., 2023; Prakash, 2019; Sorooshian et al., 2000). However, different precipitation products can readily cause a large

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impact on the accuracy of hydrologic modeling because of their data sources, retrieval algorithms, and satellite sampling (Su et al., 2021; Yin et al., 2021). Lei et al. (2022) quantified the uncertainty contributions of multiple precipitation products in hydrological simulations using the TOPMODEL hydrological model and variance decomposition methods in the Jialing River Basin, China, and showed that MSPs have spatial and temporal variability, with PXGB2 being the best performer. Yin et al. (2021) Used a Bayesian Model Averaging (BMA) based algorithm to integrate the four precipitation data and optimized the weights to obtain a fusion product that outperforms a single data source in daily statistics and seasonal analysis. Sun et al. (2018) evaluated the performance of four satellite precipitation products in the humid regions of southern China and found that TMPA performed best in capturing extreme precipitation and simulated runoff, while CMORPH combined with the VIC model was the most efficient in detecting extreme runoff. There are many current studies that used fusion algorithms to reduce errors in multi-source precipitation products along with simulation evaluation using hydrologic models (Shen et al., 2022; Stephens et al., 2022; Wu et al., 2023). These studies indicated that the differences in the performance of hydrologic modeling using different precipitation products varied across different basin and regions, mainly due to the high level of uncertainty in the data sources, inversion algorithms, and bias calibration processes of the satellite precipitation products. Uncertainty in the final satellite precipitation product coupled with parameter uncertainty and structural uncertainty in the hydrologic modeling will result in uncertainty in runoff or other modelled hydrologic fluxes.

Although many studies currently applied numerous precipitation products to hydrologic modeling, most of them are biased towards performance evaluation of hydrologic modeling of precipitation products without quantifying the causes of simulation discrepancies in terms of internal processes (Chegwidden et al., 2019; Schöniger et al., 2014). In other words, the traditional simulation metrics NSE, R^2, etc. simply answer the question of how model performance differs, but do not provide information on why hydrological models differ as they do, which is an important element of uncertainty assessment (Weijs et al., 2018). Bennett et al. (2019) quantified hydrological model behavior using transfer entropy and found that SUMMA runoff in the Snake River region is primarily driven by snowmelt, whereas VIC and PRMS are primarily influenced by precipitation and evapotranspiration, revealing differences in the transfer of information from the models at different time scales. Zhou et al. (2021) used transfer entropy to quantify information transfer between hydrological model components and found that information transfer was significantly different for different model structures and correlated with catchment characteristics, highlighting the need to move beyond approaches based solely on runoff calibration to improve model fidelity. Franzen et al. (2020) found that information on past wetness status in the Upper Colorado River Basin was the most informative for flow prediction through information theoretic methods, and that this dependence was strongest in autumn with the longest lag time in spring. The above studies all quantified the difference of model structure on the entropy of information transfer in hydrological systems from the perspective of model structure. However, there are fewer studies that quantify the process level differences under driven simulations from the perspective of different precipitation data sources. Stated differently, this means quantifying flux differences and process connectivity during hydrological modeling, rather than just assessing the performance merits of hydrological modeling.

Many studies have shown that entropy of information transfer can quantify process-level differences within model simulations, thus providing directions for hydrological modeling improvements (Goodwell et al., 2020; Ruddell et al., 2019). Therefore, the primary objective of this study was to assess the value of different precipitation data sources for hydrological simulation in inland arid basin, to drive the Variable Infiltration Capacity (VIC) hydrological model for simulation experiments using seven state-of-the-art precipitation products as well as interpolated data from meteorological stations, and to evaluate the performance of the hydrological simulation of the six main precipitation products by using DBMA fusion, and data assimilation methods to improve the accuracy of the hydrological simulation. Finally, the runoff simulation uncertainty for different precipitation data sources and model structures was quantified based on the DREAM algorithm, and the information transfer entropy used to quantify the hydrological modeling process level differences driven by different precipitation data.

The paper is structured as follows. Section 2 describes the study area, the data used and the methodology. Section 3 presents a detailed analysis of the results. Section 4 introduces the discussion followed by conclusions section 5.

2. Materials and methods

2.1. Study area

The study area, Ebinur lake basin, is in the Xinjiang Uygur Autonomous Region of China, which is geographically located in the southwestern part of the Junggar Basin and consists mainly of deserts, Gobi and grasslands (Fig. 1(a)). The region suffers from severe environmental problems, with water scarcity leading to the reduction in the size of Lake Ebinur every year, as well as problems of salinization and dust storms. The study area has a typical inland arid climate, with an average annual precipitation of 200–400 mm, high summer temperatures and low winter temperatures, with obvious seasonal differences. Soils are zonally distributed, with grey-brown and grey-desert soils dominating the plains, meadow soils and potassium-calcium soils dominating the mountainous areas, swampy soils and saline soils around the lake area, and sandy soils widely distributed in the lakeshore plains and the lakeshore deltas. Vegetation distribution presents an altitudinal band, including bands of alpine meadows, subalpine meadows, forests, grasslands, and desert steppes to deserts. According to the results of MODIS data product classification, the study area is dominated by grassland (61.27 %), bare lands (20.06 %) and agricultural lands (6.53 %), with a high percentage of 63.79 % of natural vegetation and only 7.95 % of agricultural vegetation (Fig. 1(c)).

2.2. Data sources

Meteorological data consisted mainly of station and remotely sensed reanalysis material, where the meteorological station variables were precipitation, maximum temperature, minimum temperature and wind speed, respectively. Remote sensing reanalysis data were all precipitation data (Table .1). Since HRLT, a high-precision product developed based on observational meteorological data from the China Meteorological Administration (CMA), was also used for comparison with the fused precipitation data and the original members (Qin et al., 2022). Meteorological data were selected based on the requirements of the VIC hydrological model driving inputs (Zhang et al., 2018). Also, this study used CMADS site data for interpolation (Meng et al., 2018) (Fig. 1(b)).

Soil and vegetation library parameters were obtained from the VICGlobal product data, a dataset that can be used to run VIC hydrological models on a regional to continental scale (Schaperow et al., 2021). The land cover parameters were obtained from the MODIS MCD12C1 data product, and the average land cover values for the period 2008 to 2016 were first calculated using Google Earth Engine to assign land cover types to each modeling grid cell at 500 m spatial resolution (Sulla-Menashe and Friedl, 2018).

Snow and elevation band parameters were used with the ASTER GDEM DEM data product (30m), and seven poster elevation bands were produced to account for the effect of sub grid elevation heterogeneity on hydrological flux simulations (Tachikawa et al., 2011). In addition, this study also modelled glacier changes for hydrological simulations, with glacier data from the Second Glacier Inventory of China (Guo et al., 2015).

Hydrological modeling output flux validation data including runoff

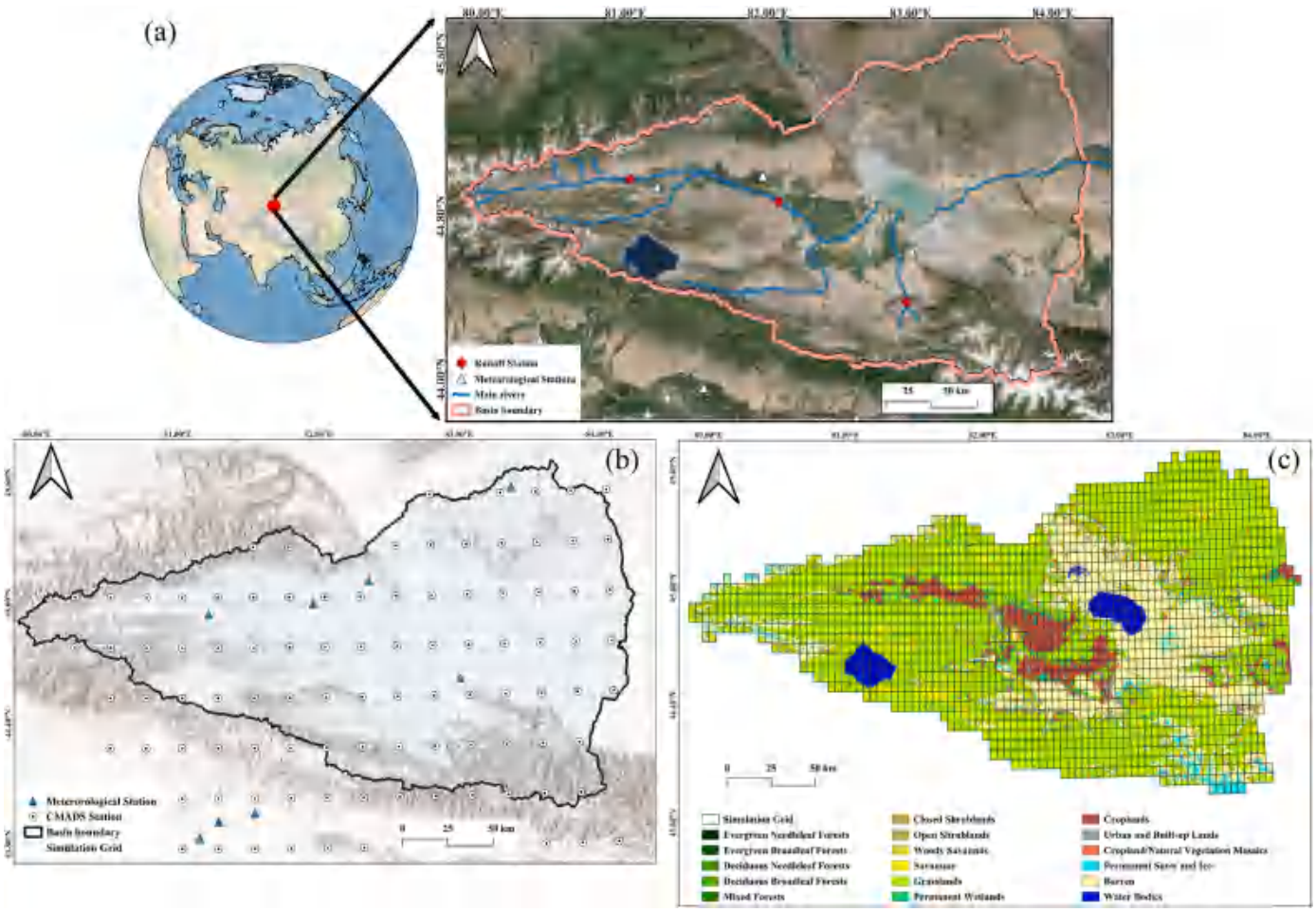

Fig. 1. Location map of the study area, schematic of the hydrological modeling grid and diagram of the land cover tiles.

(R), evapotranspiration (ET), soil moisture (SM) and snow water equivalent (SWE). Measured runoff and evapotranspiration data were obtained from local water resources and national meteorological departments, respectively, and included measured long-duration runoff from three hydrological stations. Due to the lack of measured soil moisture and snow water equivalent measured data, GLEAM and ERA5 data were used for additional assessment, respectively. GLEAM calculates potential evapotranspiration using the Priestley and Taylor equations by integrating the estimation of each component of terrestrial evapotranspiration and soil moisture, while adjusting actual evapotranspiration by combining microwave vegetation optical depth (VOD) data (Martens et al., 2017). The ERA5-Terrestrial Reanalysis dataset provides SWE observation products to obtain monthly averages to assess the accuracy of model simulations of SWE (Hersbach et al., 2020).

2.3. Methodology

In this study, daily-scale hydrological variables (runoff, evapotranspiration, surface soil moisture, and snow water equivalent) were simulated for the period 2008 to 2016 using the VIC hydrological model of the coupled glacier model in conjunction with multi-source meteorological data. Eight sensitive model parameters were calibrated and optimized by the DREAM algorithm, and an ensemble Kalman filter was used to assimilate the observations to improve the simulation accuracy. Finally, transfer entropy was used to describe the information transfer of hydrological fluxes in simulation scenarios with different data sources (Fig. 2). The following are specific descriptions of the methods used.

2.3.1. Distributed hydrological model

The Variable Infiltration Capacity (VIC) model is a distributed hydrological model used to simulate water and energy balances, including hydrological processes such as evapotranspiration, interception, surface runoff, and baseflow (Hamman et al., 2018). The VIC model operates by simulating each grid cell independently and considering vegetation heterogeneity, soils with different infiltration rates, and topographic layers, allowing fluxes to be updated at daily or sub-daily time steps (Gao et al., 2010). In combination with the routing model described by Lohmann et al. (1996) the VIC model can simulate the transport of water from the grid cell to the outlet. In addition, this study simulated glacier changes including glacier accumulation, distribution and ablation by Hamman et al. (2017) scheme based on volume-area relationship to calculate glacier distribution and snow water equivalent. The main parameters of the model were shown in Table S1&S2 in the Appendix.

2.3.2. Meteorological data interpolation and fusion

In this study, the thin plate spline function (TPS) and inverse distance weight (IDW) methods were used to spatially interpolate the meteorological data in the study area to support the meteorological driving data requirements of the VIC model. Two interpolation methods were also used to generate an interpolated dataset of weather-driven sites in the study area by combining the CMADS site and national weather site data, respectively. IDW_station, TPS_station, IDW_cmads and TPS_cmads correspond to the corresponding interpolation methods and station data, where IDW is the inverse distance weight method, TPS is the thin plate spline function method, and station and camds correspond to the national meteorological station data and CMADS station data, respectively.

Table 1
Precipitation product dataset information.

Name	Resolution	Scope	Time
CHIRPS	0.05°, daily	50°S - 50°N	1981- present
CMFD	0.1°, daily	China (land only)	1979–2018
CMORPH	0.25°, daily	60°S - 60°N	1998- present
PERSIANN-CDR	0.25°, daily	60°S - 60°N	1983-present
TRMM	0.25°, daily	50°S - 50°N	1998–2019
ERA5	0.1°, hourly	Earth	1950-present
HRLT*	1 km, day	China (land only)	1998–2018

Note: Data marked with an asterisk (*) did not participate in the precipitation data fusion. CHIRPS, short for Climate Hazards Group InfraRed Precipitation with Station data, is a high-resolution satellite precipitation product jointly released by the U.S. Geological Survey (USGS) and the Climate Hazards Group at the University of California, Santa Barbara. CMFD, short for China Meteorological Forcing Dataset, is a high-resolution gridded near-surface meteorological dataset developed by the Institute of Tibetan Plateau Research, Chinese Academy of Sciences. CMORPH, short for Climate Prediction Center Morphing Technique, is a satellite-based precipitation estimation product developed by the Climate Prediction Center (CPC) of the National Centers for Environmental Prediction (NCEP). PERSIANN-CDR, short for Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks - Climate Data Record, is a global, long-term precipitation dataset developed by the Center for Hydrometeorology and Remote Sensing (CHRS) at the University of California, Irvine. TRMM, short for Tropical Rainfall Measuring Mission, was a joint space mission by NASA and the Japan Aerospace Exploration Agency (JAXA) designed to monitor and study precipitation and energy release in tropical and subtropical regions. ERA5, short for Fifth Generation of the European Reanalysis, and ERA5 Land Surface Reanalysis, is an enhanced global dataset produced by the European Center for Medium-Range Weather Forecasts (ECMWF) as part of the land component of the fifth-generation European reanalysis. HRLT, short for High Resolution Land and Atmosphere, is a high-resolution precipitation product developed based on Chinese Meteorological data (CMA) for land and atmosphere interactions.

The Thin Plate Splines (TPS) interpolation method is a non-parametric method for interpolation and smoothing of multidimensional data (Keller and Borkowski, 2019). It generates smooth interpolated surfaces by minimizing the integral of the objective function, i.e. the curvature of the spline surface. The Inverse Distance Weighting (IDW) interpolation method is a distance-based deterministic interpolation technique commonly used in GIS and environmental sciences (Bartier and Keller, 1996). The basic principle is that the value of the point to be interpolated is calculated from the weighted average of the known points around it, with the weights being proportional to the inverse of the distance.

2.3.2.1. DBMA technique. To obtain more accurate precipitation data, this study uses Bayesian model averaging to fuse data from six satellite precipitation products (CMORPH, CHIRPS, CMFD, PERSIANN-CDR, TRMM, and ERA5) that are more widely used both domestically and internationally (Islam et al., 2020; Maggioni and Massari, 2018). Dynamic Bayesian Model Averaging (DBMA) is a statistical post-processing technique designed to integrate predictions from multiple models to improve prediction accuracy and reliability (Ma et al., 2018a). DBMA calculates the overall PDF of a mixing variable (e.g., precipitation) by combining the predictions of each model with its conditional probability density function (PDF). This process is based on the full probability formula, where the weights of each model, i.e. the posterior probabilities, are derived based on its performance during the training period. In precipitation forecasting applications, DBMA can use either a normal distribution approximation or a hybrid model combining discrete-continuous components, the latter of which determines the gamma distribution parameters through logistic regression and linear regression, thus accurately describing the conditional PDF of precipitation.

A brief description of the DBMA method is provided as follows. It is assumed that the fused precipitation variable is denoted as y, where k represents the number of ensemble members, and each ensemble member's prediction f_i is associated with a conditional probability density function $h_i(y|f_i)$, which can be interpreted as the PDF of y conditioned on f_i, where f_i is the best prediction among the ensemble members. The PDF of y is given by

$$p(y|f_1, f_2, \cdots, f_k) = \sum_{i=1}^{k} w_i h_i(y|f_i) \tag{1}$$

where $p(y|f_i)$ represents the forecast PDF of the DBMA ensemble y conditioned on the observed data; and w_i represents the relative contribution of predictor i to the predicted variable during the training period, also referred to as the posterior probability of member i. In order to guarantee that the ensemble prediction forms a complete probability distribution during the training period, the sum of the posterior probabilities of all members must equal 1.

In the standard DBMA used by Raftery et al. (2005), the conditional PDF $P(y|f_i)$ is approximated using a normal distribution, whereas in the "discrete-continuous" mixture model proposed by Sloughter et al. (2007), the conditional PDF of the precipitation variable y is expressed as follows:

$$p(y_0|f_1, f_2, \cdots, f_k) = \sum_{i=1}^{k} w_i \Big[p(y_0 = 0|f_i) I[y_0 = 0] + , \tag{2}$$

$$p(y_0 > 0|f_i) g_i(y_0|f_i) I[y_0 > 0] \Big]$$

where y_0 represents the cube root of cumulative precipitation, and the general indicator function $I[y_0 = 0]$ or $I[y_0 > 0]$ takes the value of unity if the condition in the brackets is satisfied, and zero otherwise. The term $p(y_0 = 0|f_i)$ is defined by Equation (3). The conditional PDF $g_i(y_0|f_i)$ of the precipitation amount y_0, given that it is positive, follows a gamma distribution with the corresponding PDF.

$$logitp(y_0 = 0|f_i) = a_{0i} + a_{1i} f_i^{1/3} + a_{2i}\delta_i \tag{3}$$

$$g_i(y_0|f_i) = \frac{1}{\beta_i^{\alpha_i}\Gamma(\alpha_i)} y_0^{\alpha_i - 1} \exp(-y_0 / \beta_i) \tag{4}$$

The *logit* function is used, where a_{0i}, a_{1i}, and a_{2i} are parameters determined through logistic regression, with precipitation (or zero precipitation) as the dependent variable and $f_i^{1/3}$ and δ_i as the predictor variables. If $f_i = 0$, then δ_i is set to 1; otherwise, it takes the value of 0. Additionally, Γ denotes the gamma function. The parameters $\alpha_i = \mu_i^2/\sigma_i^2$ and $\beta_i = \sigma_i^2/\mu_i$ of the gamma distribution are influenced by f_i through the relationships given below:

$$\mu_i = b_{0i} + b_{1i} f_i^{1/3} \tag{5}$$

$$\sigma_i^2 = c_0 + c_1 f_i \tag{6}$$

where μ_i and σ_i^2 represent the mean and variance of the distribution, respectively. The parameters b_{0i} and b_{1i} are obtained through linear regression, while c_0 and c_1 are estimated using the maximum likelihood method based on the training data.

The model parameters are estimated based on the training period data, where the training period refers to the simulation and observation data from the initial N days. Different training periods were chosen for weight estimation; for instance, Sloughter et al. (2007) selected a 30-day training period to apply BMA for 48-h daily forecasts in the North Pacific region, while Ma et al. (2018a) proposed a DBMA algorithm with a 40-day training period to merge multiple satellite precipitation datasets for the Tibetan Plateau region. It has been demonstrated that the choice of training period is highly dependent on the specific conditions of the study area, and validation using actual data from the region is considered more reliable.

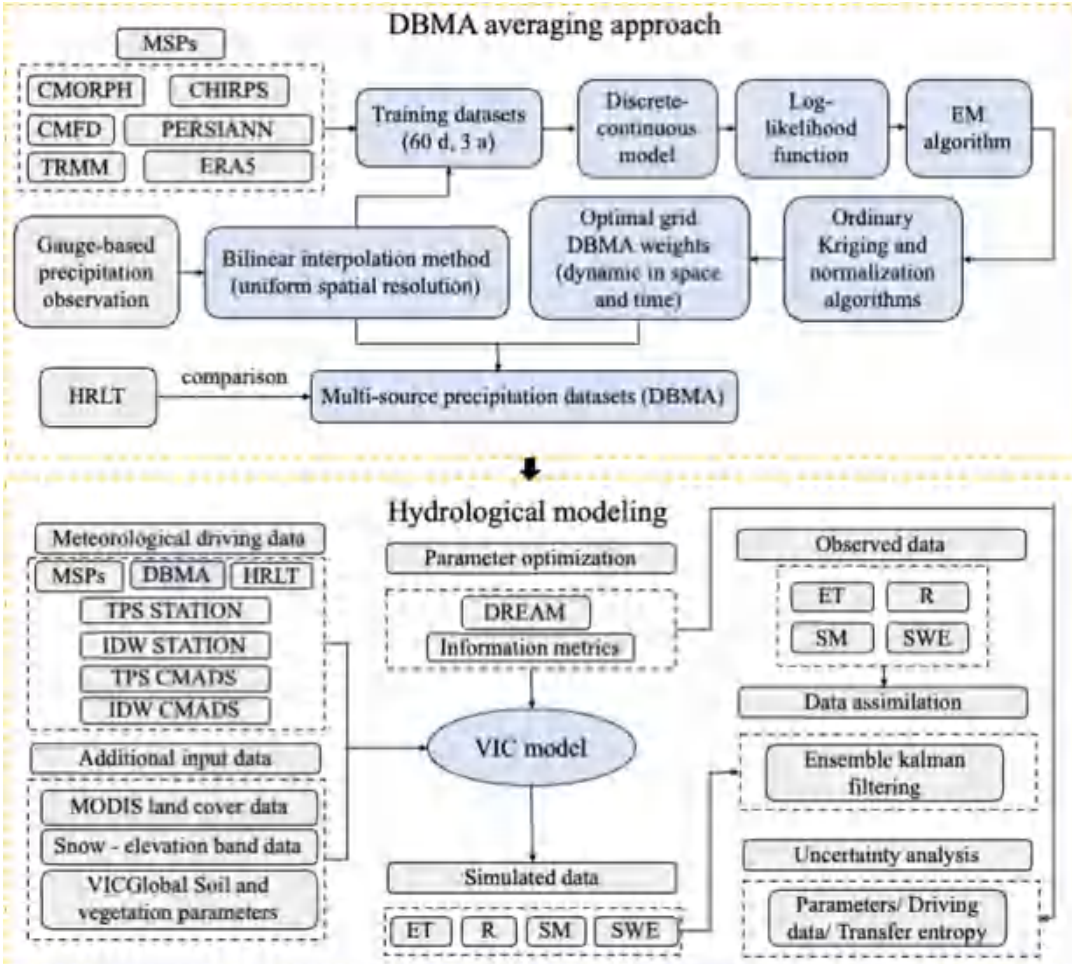

Fig. 2. Methodological framework.

The model training length was assessed by computing the root mean square error (RMSE) and Pearson's correlation coefficient (CC) for the DBMA dataset, using precipitation data from 4 meteorological stations in study area during 2016, and comparing these metrics across various potential training periods (N = 20, 25, 30, …, 100). Fig. 3 showed that as the training period length increased, the RMSE and CC exhibited fluctuating decreases (increases), with both reaching their optimal values at 60 days. After 60 days, with further increases in the training period, the RMSE and CC respectively increased and decreased, yielding suboptimal results. Hamill (2007) suggested that extending the training period to include the same period of the same season from previous years might enhance the performance of the DBMA model. Therefore, in this study, a 60-day training period was adopted, and the training period was optimized by selecting a total of 180 days (three 60-day periods, denoted as 3a) from the same time of the current year and the previous two years to optimize the DBMA weights (see Fig. 3).

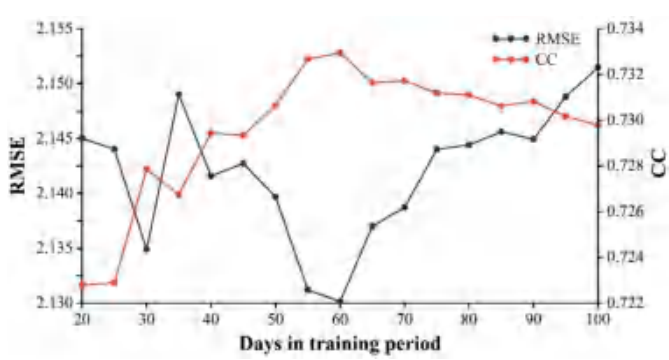

Fig. 3. Variations in RMSE and CC between DBMA-based precipitation estimates and ground-based observational data across different training period lengths. RMSE: root mean square error; CC: Pearson's correlation coefficient.

2.3.3. *Data assimilation and parameter optimization algorithms*

2.3.3.1. Ensemble kalman filter. Data assimilation is a method of combining observed data with model predictions for improving the predictive accuracy of the system (Houtekamer and Mitchell, 1998). Data assimilation was performed by using Ensemble Kalman Filtering (EnKF) in this study, which is a Monte Carlo approximation of Kalman Filtering that calculates the statistical characteristics of a future state by generating a series of random variables to represent the probability distribution of the system's state (Katzfuss et al., 2016). EnKF overcomes the limitation of the traditional assumption of constant background error variance by adding random perturbations to the set of states from observations. The process consists of an initial background field, predicting the state values at the next moment, calculating the Kalman gain matrix, updating the mean and covariance of the state variables, and repeating the process for successive time steps. The advantage of EnKF lies in its adaptability to nonlinear systems without the need for tangential and concomitant modes of the forecast operator, making it effective in dealing with nonlinear and discontinuous problems in dynamic models. This approach enables more accurate prediction of hydrological fluxes and improves the simulation accuracy and reliability of

hydrological models.

The EnKF leverages the Monte Carlo method by representing the probability density function of the random dynamic forecast with a set of random variables (N) that follow a Gaussian distribution, and through forward integration, the probability density function of the state ensemble at the next time step is computed, thus obtaining its statistical characteristics and avoiding the assumption that the background error variance is known or remains constant over time. In the EnKF, the observation ensemble is generated by adding Gaussian disturbances with a mean of 0 and a variance equal to the observational error to the observed values; otherwise, the updated ensemble would result in an excessively small covariance. The computational steps of the EnKF are outlined as follows.

(a) The background field is initialized by generating N random variables X_i (i = 1, …, N) that followed a Gaussian distribution at time step k, where the state variables are the observed hydrological variables, including runoff, evapotranspiration, soil moisture, and snow water equivalent.

(b) The forecast values of the random variables at time k + 1, X_i^{k+1}, were calculated, with the nonlinear system expressed through discretized state and observation equations, allowing the forecast values at k + 1 to be computed as follows:

$$X_i^{k+1} = f\left(X_i^k\right) + \omega(i,k), \omega(i,k) \sim N(0, Q_k) \tag{7}$$

In the above equation, X_i^{k+1} represents the forecast value at time k + 1; $f\left(X_i^k\right)$ denotes the process model, which in this study corresponds to the VIC model; $\omega(i,k)$ represents the uncertainty of X_i^{k+1}; Q_k is the model error matrix; and $\omega(i,k)$ is Gaussian white noise with a mean of 0 and a variance of Q_k.

(c) The Kalman gain matrix K_{k+1} at time k + 1 is computed as follows:

$$K_{k+1} = P_{k+1}^{-} H^T \left(HP_{k+1}^{-} H^T + R_{k+1}\right)^{-} = C_{k+1}^{xy}\left[C_{k+1}^{xy} + R_{k+1}\right]^{-} \tag{8}$$

In the above equation, H represents the linear observation transformation matrix, which is set to 1 in this study; R_{k+1} is the observation error; $HP_{k+1}^{-} = C_{k+1}^{xy}$ represents the variance of the model predictions; and P_{k+1}^{-} denotes the error covariance matrix of the state variable ensemble at time k+1, indicating the covariance between the model state variables and the model predictions at time k+1. The error covariance matrix P_{k+1}^{-} is expressed by the following relationship:

$$P_{k+1}^{-} = E\left[e_{k+1}^{-} {e_{k+1}^{-}}^T\right] = {}^1/_{n-1} e_{k+1}^{-} {e_{k+1}^{-}}^T \tag{9}$$

$$e_{k+1}^{i-} = \left\{x_{k+1}^{i} - x_{k+1}^{-}\right\} \quad i = 1, 2, \cdots, N$$

$$e_{k+1}^{-} = \left\{e_{k+1}^{1-}, e_{k+1}^{2-}, \cdots, e_{k+1}^{N-}\right\} P_{k+1}^{-} H^T = C_{k+1}^{xy}$$

(d) The state variable average X_{k+1}^{a} at time k+1k+1k+1 is calculated by using the Kalman gain matrix, and the individual state variables $X_{k+1}^{a,i}$ are then updated accordingly.

$$X_{i,k+1}^{a} = X_i^{k+1} + K_{k+1}\left(O_{k+1} - HX_i^{k+1} + \gamma_{i,k+1}\right), \tag{10}$$

$$\gamma_{i,k+1} \sim N(0, R_{k+1})$$

$$\overline{X}_{k+1}^{a} = \frac{1}{N}\sum_{i=1}^{N} X_{i,k+1}^{a} P_{k+1}^{a} = \frac{1}{N-1}\sum_{i=1}^{N}\left(X_{i,k+1}^{a} - \overline{X}_{k+1}^{a}\right)\left(X_{i,k+1}^{a} - \overline{X}_{k+1}^{a}\right)^T$$

In the above equation, O_{k+1} represents the observation at time k+1, and $\gamma_{i,k+1}$ denotes Gaussian white noise with a mean of 0 and variance R_{k+1} at time k+1.

(e) The calculation proceeds to the next time step, returning to step (b).

2.3.3.2. Parameter optimization and performance evaluation metrics. The DREAM (Differential Evolution Adaptive Metropolis) algorithm is a global optimization algorithm for Bayesian parameter estimation, mainly applied to Markov Chain Monte Carlo (MCMC) methods (Zahmatkesh et al., 2015). It improves sampling efficiency and exploration by combining Differential Evolution and the adaptive Metropolis algorithm (Vrugt et al., 2009). The central concept of the DREAM algorithm is to search in the parameter space using multiple parallel chains, each of which is updated based on the state of the others, thus enhancing the global search capability and avoiding the trap of local extremes.

Eight widely used metrics were employed in this study to assess the performance of fused precipitation data, original ensemble members, meteorological interpolation data, and hydrological model output variables. The performance was evaluated using eight widely used metrics: Nash-Sutcliffe Efficiency (NSE), Root Mean Square Error (RMSE), Coefficient of Determination (R^2), Kling-Gupta Efficiency (KGE), Pearson Correlation Coefficient (R), Probability of Detection (POD), False Alarm Rate (FAR), and Critical success index (CSI), with the following formulas provided:

$$NSE = 1 - \frac{\sum_{t=1}^{n}\left(Q_{obs}(t) - Q_{sim}(t)\right)^2}{\sum_{t=1}^{n}\left(Q_{obs}(t) - \overline{Q_{obs}}\right)^2} \tag{11}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(Q_{obs}(t) - Q_{sim}(t)\right)^2} \tag{12}$$

$$R^2 = \frac{\left(\sum_{t=1}^{n}\left(Q_{obs}(t) - \overline{Q_{obs}}\right)\left(Q_{sim}(t) - \overline{Q_{sim}}\right)\right)^2}{\sum_{t=1}^{n}\left(Q_{obs}(t) - \overline{Q_{obs}}\right)^2 \sum_{t=1}^{n}\left(Q_{sim}(t) - \overline{Q_{sim}}\right)^2} \tag{13}$$

$$KGE = 1 - \sqrt{\left(\frac{\sigma_{sim}}{\sigma_{obs}} - 1\right)^2 + \left(\frac{\mu_{sim}}{\mu_{obs}} - 1\right)^2 + \left(\frac{\rho_{sim,obs}}{1} - 1\right)^2} \tag{14}$$

$$R = \frac{\sum_{t=1}^{n}\left(Q_{obs}(t) - \overline{Q_{obs}}\right)\left(Q_{sim}(t) - \overline{Q_{sim}}\right)}{\sqrt{\sum_{t=1}^{n}\left(Q_{obs}(t) - \overline{Q_{obs}}\right)^2 \sum_{t=1}^{n}\left(Q_{sim}(t) - \overline{Q_{sim}}\right)^2}} \tag{15}$$

$$POD = \frac{TP}{TP + FN} \tag{16}$$

$$FAR = \frac{FP}{FP + TP} \tag{17}$$

$$CSI = \frac{TP}{TP + FP + FN} \tag{18}$$

where $Q_{obs}(t)$ represents the observed value, $Q_{sim}(t)$ represents the simulated value, and $\overline{Q_{obs}}$ denotes the mean of the observed values; where μ is the mean, σ is the standard deviation, and $\rho_{sim,obs}$ is the correlation coefficient between the simulated and observed values; where TP is the number of true positives, FN is the number of false negatives, and FP is the number of false positives.

The optimal value ranges for the metrics are as follows: NSE should be between 1 and $-\infty$, with values closer to 1 indicating better model

performance; RMSE should be as low as possible, with 0 representing perfect accuracy; R^2 should range from 0 to 1, with 1 indicating perfect correlation between simulated and observed values; KGE ranges from $-\infty$ to 1, with 1 representing perfect efficiency; R ranges from −1 to 1, with values closer to 1 indicating better correlation; POD ranges from 0 to 1, with 1 representing perfect detection; FAR should be as low as possible, with 0 representing no false alarms; and CSI ranges from 0 to 1, with 1 indicating perfect prediction accuracy (Ma et al., 2018b; Wei et al., 2023; Yang et al., 2024).

2.3.4. Information metrics approach

The foundations of information theory were proposed by Shannon (1931), introducing the concept of information entropy to describe the uncertainty of a continuous random variable X, defined using the probability density function p(x). For multivariate systems, the joint entropy H(X, Y) takes into account the common distribution of variables X and Y. Mutual information I(X; Y) quantifies the sharing of information between X and Y, while conditional entropy H(X|Y) and conditional mutual information I(X; Y|Z) extend these concepts to measures of information given other variables. The transfer entropy T(X→Y) introduced by Schreiber (2000) was calculated by conditional mutual information to quantify the transfer of information between variables, taking into account the effect of variable history. To reduce the computational complexity, a parametric approach that restricts the size of the history window and assumes the Markovian nature of the system is usually used and simplified as:

$$T_{X\to Y} = I(Y_t; X_{t-1} | Y_{t-1}) \tag{19}$$

The unit of information is nats, based on natural logarithms. The actual measurement of information entropy needs to rely on estimation methods, including parametric and non-parametric techniques. In this study, the k-nearest-neighbor method is used for non-parametric estimation, which outperforms the traditional histogram method, especially when dealing with large and high-dimensional datasets (Kataria and Singh, 2013). This method does not assume a specific density distribution but requires a large amount of data to ensure the accuracy of the estimate. Estimates of entropy, mutual information, conditional mutual information and transfer entropy are based on distance to the kth nearest neighbor measure. By refining the methodology of these estimates, this study aims to minimize the average bias in information entropy. Ultimately, by visualizing the information transfer network as a chordal graph, it is possible to identify coupling between variables and

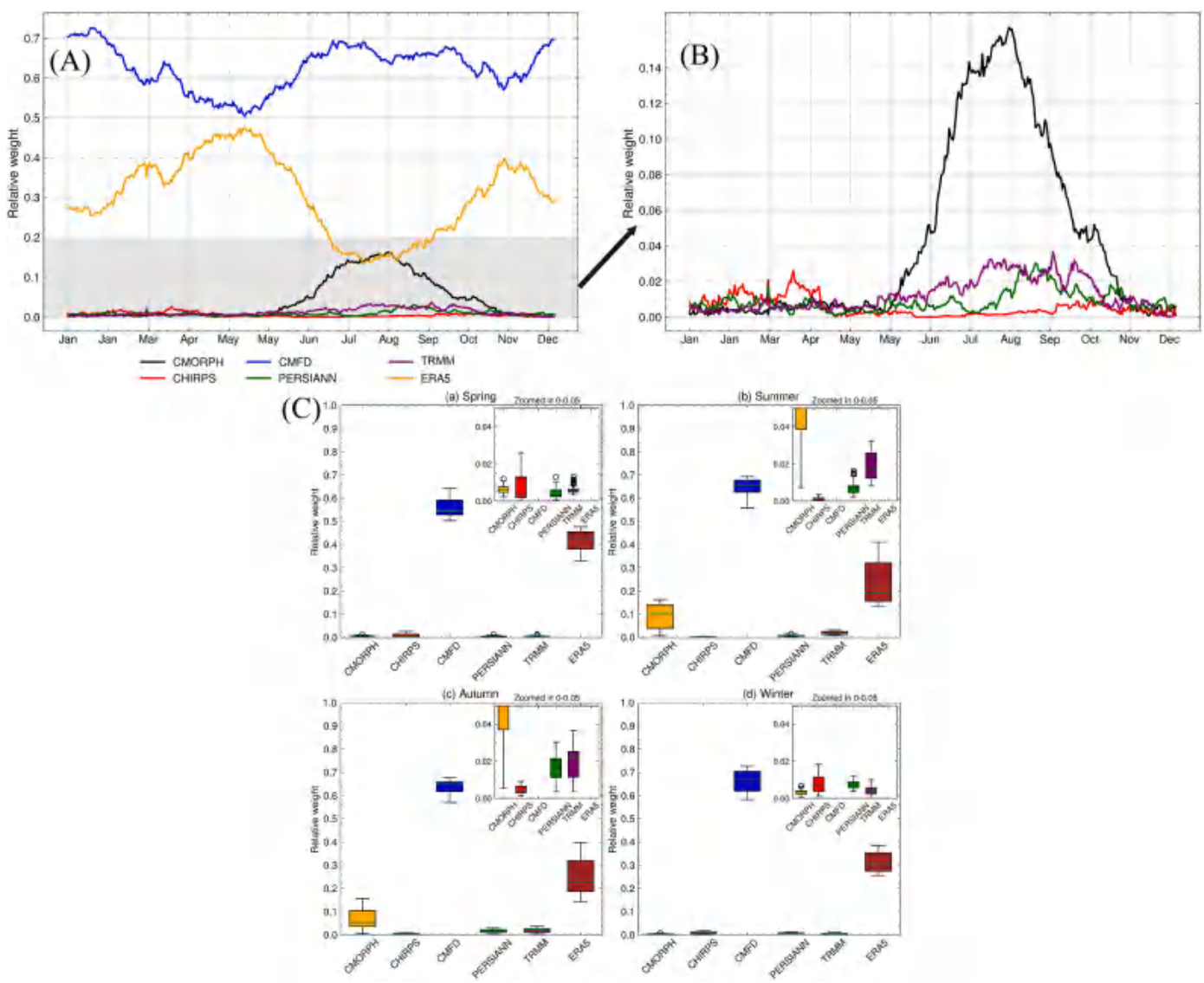

Fig. 4. Weight distributions from DBMA fusion of precipitation data (A and B are multi-year average weight distributions, and C is the average relative weights for different seasons).

understand the network structure of the information flow, although the outer arc length synthesis is not a direct reflection of the total amount of information transferred because of the inclusion of potentially redundant or synergistic components. Please refer to Bennett et al. (2019) for details.

3. Results

3.1. Analysis of DBMA precipitation data fusion results

The DBMA fusion precipitation data merged member-year average weights showed that CMFD and ERA5 had relatively high weight assignments, with their average weights being 0.57 and 0.38, respectively; the rest of the data had relatively low weight assignments. Seasonal differences in the weight distribution of different precipitation data were more obvious, among which CMFD and ERA5 had basically the same weight distribution in winter and spring seasons and showed larger differences in summer and autumn seasons; CMORPH showed a typical pattern of weight distribution in dry seasons, and the rest of the data had a gentler seasonal distribution of weights (Fig. 4).

In terms of the average relative weight distribution of precipitation products in different seasons, CMFD and ERA5 also received relatively high weights, while the proportion of allocation in different seasons was basically the same, with the sum of the two accounting for about 90 % of the total, while the rest of the products accounted for the sum of the weights was only about 10 %. In the spring and winter, the average relative weight distribution of CMFD ranged from 0.02 to 0.09, while the ERA5 ranged from 0.10 to 0.95, and the rest of the product weight distributions were basically below 0.2. In summer and autumn, CMFD weight distributions ranged from 0.01 to 0.90, while ERA-Land ranged from 0.02 to 0.80. CMFD products had relatively high weights compared to spring and winter, and CHIRPS received lower weight allocations in summer and autumn (Fig. 4).

The annual spatial variation from the DBMA fused precipitation data and the original precipitation members demonstrated that the fused data significantly enhanced spatial resolution accuracy, particularly compared to HRLT, as it effectively captured the spatial trend of precipitation-altitude gradient differentiation, with higher precipitation in surrounding high-altitude mountainous areas than in the central plains (Fig. 5). The interannual variation trend showed that the fluctuation range of the DBMA fused precipitation data was consistent with station observations (0–120 mm), although differences in fluctuation frequency were observed. Compared to other original precipitation members, the fused precipitation data significantly reduced overestimation and underestimation, as ERA5 and CMORPH data were found to underestimate observed precipitation, while PERSIANN exhibited a tendency for overestimation. The intra-annual variation trend revealed that the fused precipitation data were overestimated from February to June compared to observed precipitation, while from July to December, the trend largely aligned with observations, albeit with slight overestimation. Compared to other original precipitation members, the fused precipitation data significantly addressed the excessive overestimation observed in PERSIANN and the irregular underestimation exhibited by datasets such as ERA5.

The density scatterplot and performance metrics indicated that the fused precipitation data achieved RMSE, R^2, and BIAS values of 1.46, 0.75, and 6.00, respectively, and were predominantly distributed within the low-value range along the 1:1 regression line, demonstrating improved accuracy (Fig. 6). Compared to the original precipitation members, the fused precipitation data significantly improved the fit with observed precipitation, as evidenced by R^2 values below 0.3 for datasets such as ERA5 and TRMM. The RMSE was reduced from the highest value of 2.20 in CHIRPS to below 1.50, lower than HRLT's value of 1.64, whereas the BIAS was not significantly reduced. From the precipitation monitoring indicators (Table .2), the CSI index of the fused precipitation data was 0.54, showing no significant improvement compared to the original precipitation data, and was similar to PERSIANN and CMFD; it is worth noting that HRLT was also relatively low. The fused precipitation data performed well in terms of POD and FAE indicators, with the former reaching 0.95 and the latter 0.51. However, the original members CMFD and ERA5 also showed very high POD values, although their FAR indices were relatively low, with HRLT data exhibiting the best performance. To more clearly observe the monthly variation trends of each indicator, the monthly CSI index was calculated, revealing that the precipitation data fused using DBMA exhibited a more uniform distribution throughout the year compared to the original members, reducing instances where certain months showed relatively low CSI values, such as the lower CSI values observed in CMORPH data from January to March and November to December (Fig. S1).

3.2. Performance analysis of modelled hydrological fluxes

The hydrological fluxes simulated output in this study were runoff, evapotranspiration, soil moisture and snow water equivalent. After assimilation with the DREAM optimization algorithm and the ensemble Kalman filter, all simulated hydrological fluxes achieved a relatively well-fitted accuracy. For reasons of size, the following analysis focused on the runoff simulation performance of the Jinghe hydrological station, which was less exposed to human activities, and the rest were detailed in the appendix (Fig. S2~S6&Table. S4~S7). Before data assimilation, the performance of the model motivated by CHIRPS and TRMM precipitation products in the ordinary mode had NSEs of 0.18 and 0.15, and RMSEs of 10.94 and 11.15, respectively, which performed better relative to the other data (Table 3). The relatively worse performing driver data were CDR and ERA5 with NSEs of −1.74 and −1.20 and RMSEs of 20.00 and 17.94 respectively. Comparing to the glacier model, HRLT and CMFD data with NSE of 0.77 and 0.73 and RMSE of 5.71 and 6.26 had relatively high performance, while DBMA fused precipitation was instead poor, with PERSIANN and IDW STATION being the least effective. The overall improvement in simulated runoff performance in glacier mode was more significant (Table 4). After data assimilation, the NSEs all reached a confidence level of 0.80 or more, with relatively high NSEs of 0.99, 0.95 and 0.93 for CMFD, TPS CMADS and DBMA data, respectively.

Fig. 7showed hydrographs of simulated runoff and observed values at the Jinghe hydrological station. In terms of interannual variability, the DBMA, CHIRPS, CMFD, and HRLT-driven simulated runoff largely approximated the fluctuating trends in interannual variability of measured runoff, with PERSIANN, CMORPH, ERA5, IDW CMADS, IDW STATION, and TPS CMADS performing relatively poorly and having no significant seasonal fluctuating trends in variability. After the glacial model simulation, the seasonal fluctuation patterns of runoff simulated by almost all driving data were more pronounced, and the peak and low flows were close to each other and the measured values. This was mainly since the Jinghe Hydrological Station is located in the mountain pass, which is less affected by human activities, while glacial meltwater is the main source of recharge. From the perspective of intra-annual variation, the seasonal distribution pattern of runoff distribution was obvious, and the dry and wet seasons were summer and winter, respectively, with a high runoff value of 35 m^3/s (July) and a low value of 2 m^3/s (March). Relatively close to the measured runoff distribution in the ordinary mode were CHIRPS, HRLT, and TPS STATION; in the glacial mode, the best results were from CHIRPS, and the rest of the data were overestimated in the presence of summer runoff. All the simulated data showed a good degree of agreement after data assimilation, with DBMA and CMFD performing the best.

The cumulative curve diagram of the runoff process indicated that the measured runoff was in a rapid rise phase from 2 to 10 m^3/s, after which it was in a steady rise phase from 10 to 44 m^3/s, with 62 % of the runoff in the rapid rise phase. After the glacial model, the agreement between the simulated runoff and the measured values was greatly improved, with TRMM, DBMA, CHIRPS and HRLT being relatively good.

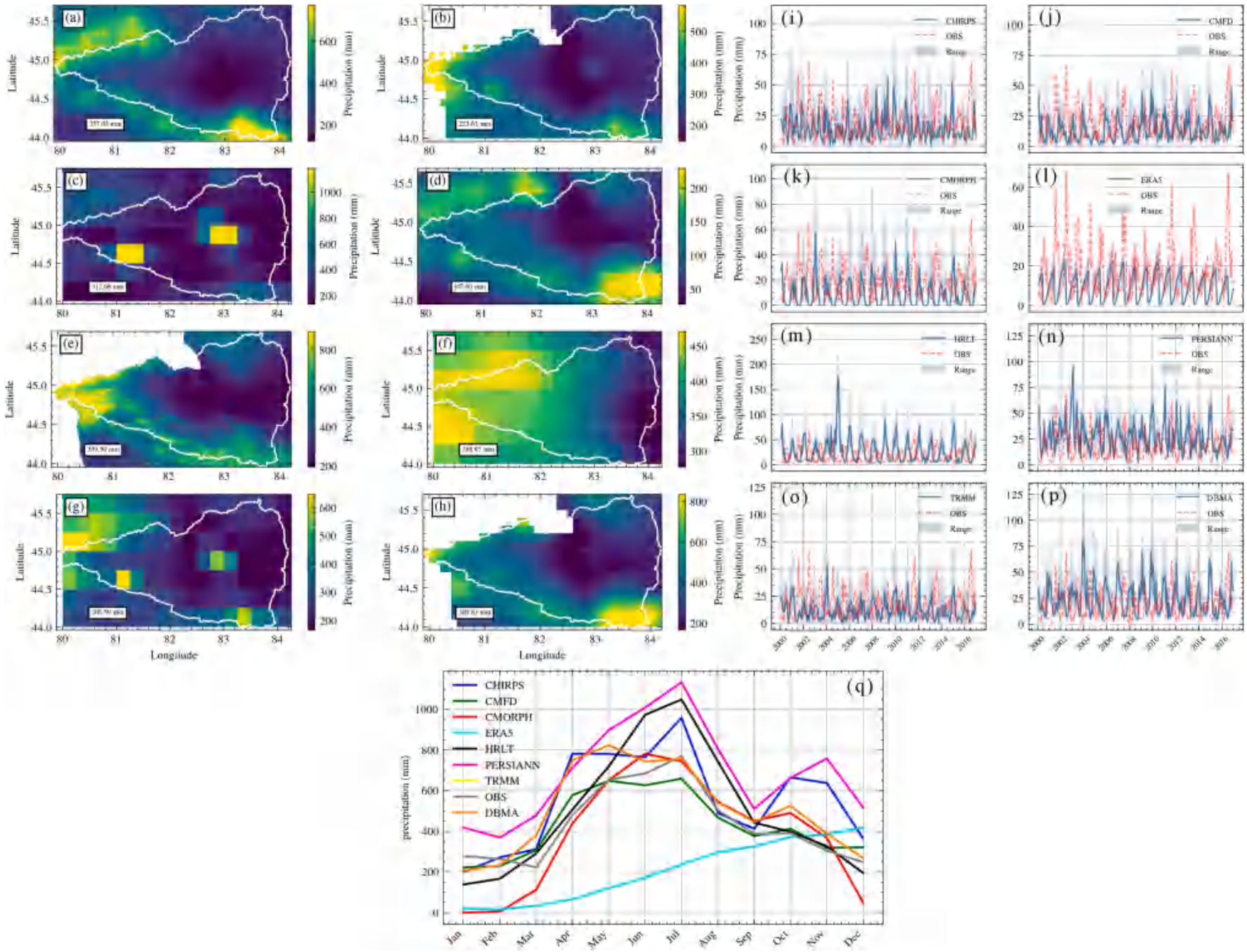

Fig. 5. Spatial and temporal variability of DBMA fused precipitation data (a, b, c, d and e are annual mean, winter, summer, spring and autumn spatial distributions, respectively; f and g are inter- and intra-annual variability, respectively).

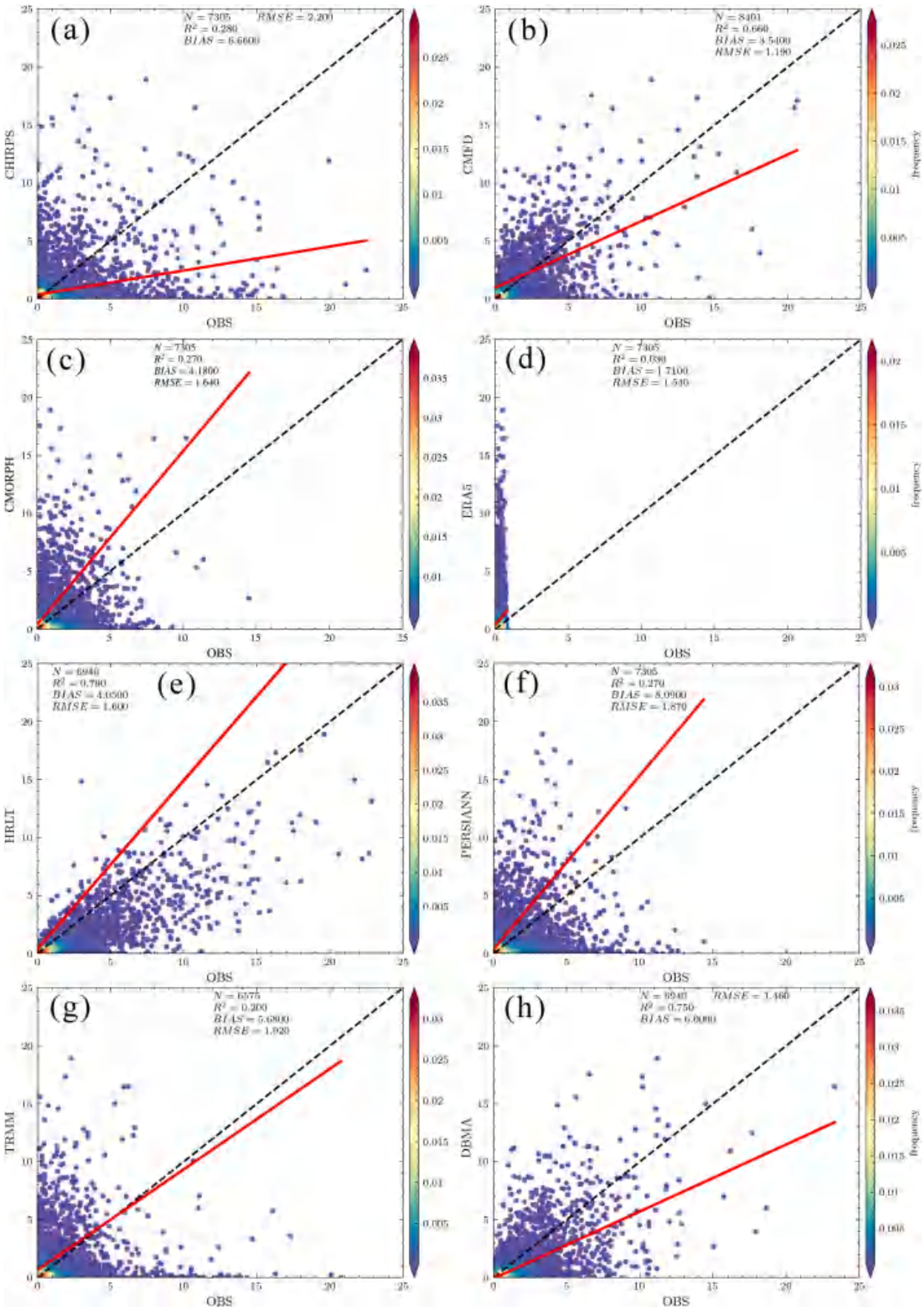

Fig. 6. DBMA fusion precipitation data validation (Subfigures a to h represent precipitation data from CHIRPS, CMFD, CMORPH, ERA5, HRLT, PERSIANN, TRMM, and DBMA, respectively.).

Table 2
Precipitation monitoring capability indicators of fused data and original members.

	CHIRPS	CMORPH	CMFD	PERSIANN	TRMM	ERA5	HRLT	DBMA
POD	0.79	0.73	0.98	0.96	0.77	0.98	0.99	0.95
FAR	0.48	0.20	0.11	0.12	0.17	0.10	0.05	0.51
CSI	0.51	0.42	0.54	0.54	0.44	0.53	0.53	0.54

Table 3
Performance assessment of runoff simulation at Jinghe hydrological station in Ebinur Lake area (monthly scale).

Name	NSE(NSEKF)		RMSE(RMSEKF)		BIAS(BIASKF)		R^2(R^2KF)		KGE(KGEKF)		R(RKF)	
PERSIANN	−1.74	0.85	20.00	4.53	−2.27	−0.02	0.00	0.87	−0.01	0.81	0.06	0.93
CHIRPS	0.18	0.92	10.94	3.30	1.58	−0.01	0.19	0.93	0.29	0.87	0.44	0.96
CMFD	0.02	0.99	11.95	0.56	−0.71	−0.00	0.04	0.99	−0.01	0.98	0.21	0.99
CMORPH	−0.21	0.89	13.28	3.89	−5.40	−0.01	0.01	0.90	−0.01	0.84	0.09	0.95
ERA5	−1.20	0.91	17.94	3.65	−0.46	−0.01	7.26	0.91	−0.01	0.85	−0.01	0.95
HRLT	−0.09	0.87	12.65	4.23	0.16	−0.01	0.08	0.88	0.24	0.82	0.29	0.94
IDW CMADS	−0.05	0.86	12.39	4.49	−1.38	−0.02	0.00	0.87	−0.23	0.81	0.02	0.93
IDW STATION	−0.43	0.87	14.42	4.32	−2.39	−0.01	0.00	0.88	−0.10	0.82	−0.03	0.94
TPS CMADS	−0.32	0.95	13.89	2.47	−4.42	−0.00	0.00	0.96	−0.04	0.91	0.03	0.98
TPS STATION	−0.15	0.84	12.94	4.75	−2.69	−0.02	0.08	0.85	0.24	0.79	0.28	0.93
TRMM	0.15	0.89	11.15	3.89	−3.48	−0.01	0.15	0.90	0.08	0.84	0.39	0.95
DBMA	0.08	0.93	11.55	3.07	−2.10	−0.00	0.11	0.94	0.17	0.88	0.34	0.97

Table 4
Performance assessment of runoff simulation at Jinghe hydrological station in the Ebinur Lake area (monthly scale, glacial model).

Name	NSE		RMSE	BIAS	R^2	KGE	R
PERSIANN	−2.27	21.84	−1.75	0.28	−0.22	0.53	−2.27
CHIRPS	0.71	6.45	9.74	0.75	0.85	0.86	0.71
CMFD	0.73	6.26	8.67	0.78	0.83	0.88	0.73
CMORPH	−0.28	13.68	−2.82	0.43	0.39	0.65	−0.28
ERA5	−0.32	13.86	−6.25	0.17	0.41	0.42	−0.32
HRLT	0.77	5.71	−4.15	0.82	0.85	0.90	0.77
IDW CMADS	−0.11	12.71	−8.22	0.04	0.09	0.19	−0.11
IDW STATION	−0.56	15.11	2.09	0.02	0.13	0.13	−0.56
TPS CMADS	0.24	10.52	−3.41	0.56	0.59	0.74	0.24
TPS STATION	0.39	9.41	−3.65	0.62	0.65	0.79	0.39
TRMM	0.64	7.15	9.45	0.73	0.79	0.85	0.64
DBMA	0.04	11.79	−4.52	0.32	0.55	0.56	0.04

With data assimilation, all data generally matched measured runoff trends, with DBMA and CMORPH having the best results.

Before data assimilation, the performance of the fused precipitation products in regional runoff simulation was found to be unsatisfactory. Specifically, for the Jinghe station, R^2, NSE, and RMSE values for the fused precipitation products were reported as 0.002, 0.08, and 11.55, respectively, indicating poor performance in runoff simulation. However, after data assimilation, a significant improvement in accuracy was achieved. These results suggested that the fused precipitation data, in the absence of assimilation, did not provide sufficient accuracy for hydrological modeling in this region. It should also be noted that most of the original ensemble precipitation members did not achieve satisfactory accuracy prior to assimilation, except for CHIRPS data. Interestingly, HRLT and TPS STATION data were found to outperform other datasets in terms of accuracy, even without assimilation, suggesting that these sources might have been more appropriate for runoff modeling in this region. Furthermore, when assessed using the glacier model, all datasets demonstrated improved performance. For example, at the Jinghe station, HRLT data yielded NSE and KGE values of 0.77 and 0.90, respectively. This indicated that the combination of high-precision precipitation data with a model structure suited to arid regions could have significantly enhanced hydrological simulation accuracy.

3.3. Comparative spatial and temporal analyses of modelled hydrological fluxes

The spatial distribution of mean annual runoff showed that the DBMA, CHIRPS and HRLT simulations exhibited higher runoff values in the southeastern and western regions, with mean values of 53.22 mm, 50.83 mm, and 65.54 mm, respectively (Fig. 8). In contrast, the PERSIANN and ERA5 simulations showed lower mean annual runoff with mean values of 16.89 mm and 12.82 mm, respectively. CMORPH modelled values presented extreme high values in some areas, reaching more than 800 mm. Seasonal spatial analysis showed that runoff was highest in summer, lowest in winter, and intermediate in spring and autumn. The DBMA, CHIRPS, CMFD, and HRLT simulations all showed high values in the summer, especially in the southeastern mountainous region and the western region, while the PERSIANN and ERA5 simulations showed low runoff values in the summer. All data-driven simulated values were generally lower in winter. The runoff in the surrounding mountains was significantly higher than the rest of the area in the glacial model, reflecting the important contribution of glacial meltwater to runoff. The simulated values of DBMA, CHIRPS, CMFD and HRLT were relatively high with mean values of 286.94 mm, 276.09 mm, 243.28 mm and 283.70 mm respectively, while the simulated values of runoff in the mountainous areas were more than 4000 mm. In contrast, the remaining plains were below 500 mm. The ERA5 and CMORPH simulations were relatively low, and their values were only above 300 mm in the mountainous areas. The interpolated data modelled runoff volumes were relatively low overall.

Seasonal water balance analyses showed that the DBMA simulated water flux balance values were relatively high in the spring (175 mm/season), with evapotranspiration and snow water equivalent changes accounting for the largest portion of the output. Simulated summer equilibrium value of 170 mm/season, with evapotranspiration accounting for more than 90 % of the output (Fig. 9). The modelled equilibrium values were relatively high in winter (70 mm/season), and the output consisted of changes in snow water equivalent and a small amount of runoff. The simulated balance value was reduced to 85 mm/season in autumn, with evapotranspiration accounting for more than half of the output. In contrast, the ERA5 simulated equilibrium values were the lowest in all seasons, with only 11.01 mm in winter, and the HRLT and CMFD simulated balances showed relatively high performance in spring and summer. In summary, during the winter months, the DBMA, CMFD, and PERSIANN data simulations showed relatively

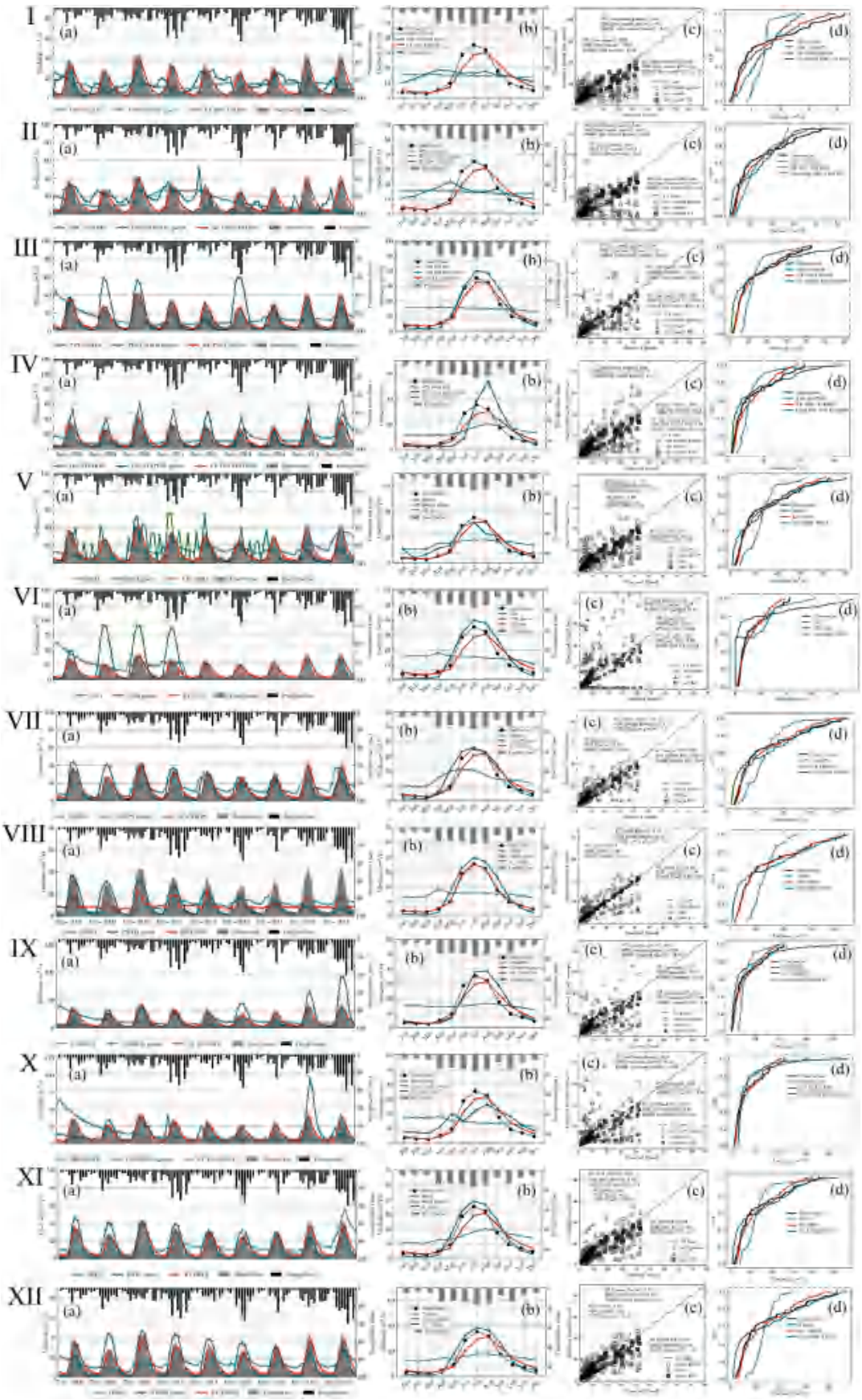

Fig. 7. Runoff calibration validation results for the Jinghe Hydrological Station (I to XII for IDW CMADS, IDW STATION, TPS CMADS, TPS STATION, DBMA, PERSIANN, CHIRPS, CMFD, CMORPH, ERA5, HRLT, and TRMM, respectively. a, b, c and d are the corresponding inter-annual variability, intra-annual variability, scatterplots and cumulative curves of runoff processes, respectively.).

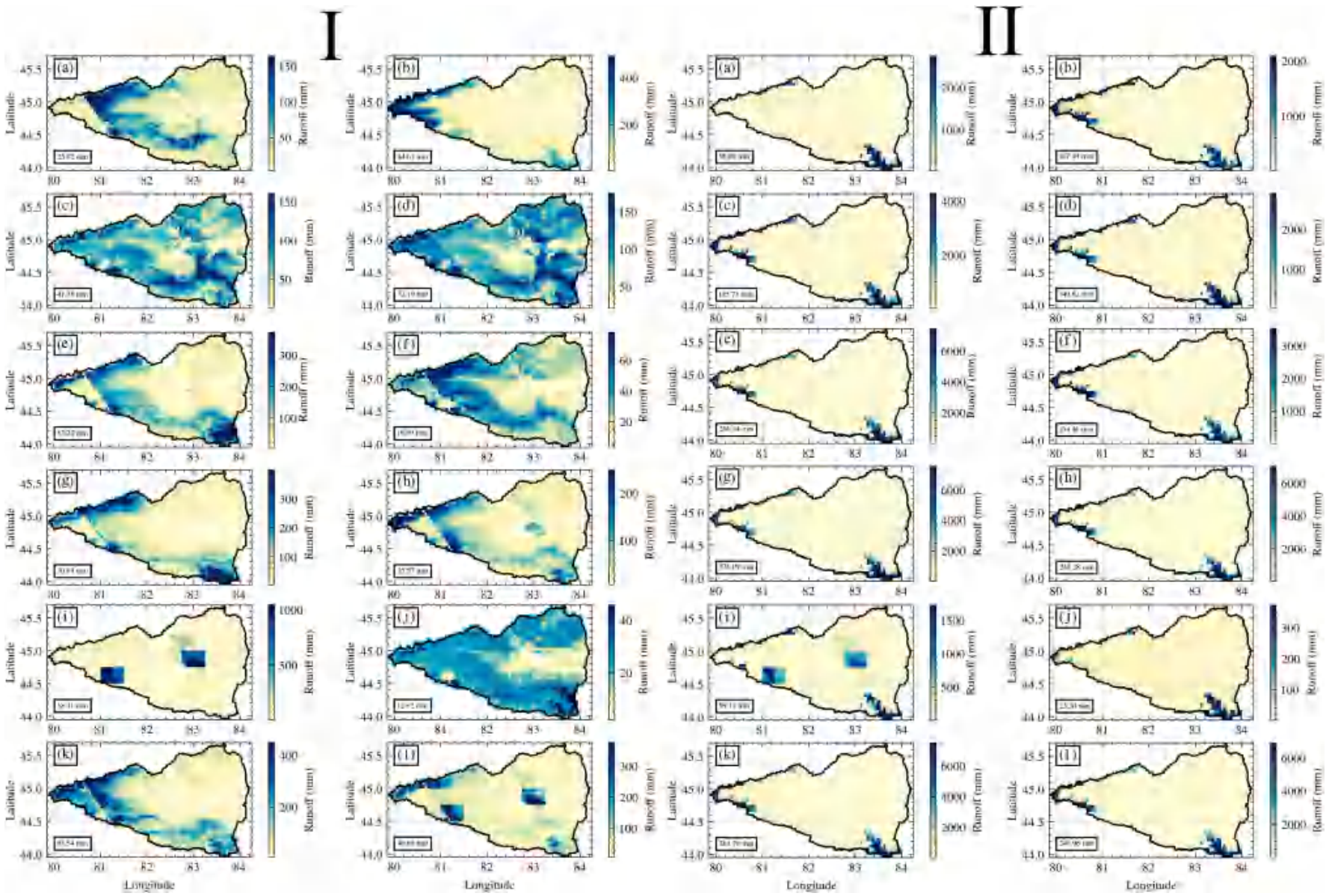

Fig. 8. Spatial comparison of annual mean simulated runoff in the study area (a, b, c, d, e, f, g, h, i, j, k and l for TPS CMADS, TPS STATION, IDW CMADS, IDW STATION, DBMA, PERSIANN, CHIRPS, CMFD, CMORPH, ERA5, HRLT and TRMM, respectively. I and II represent normal mode and glacier mode respectively).

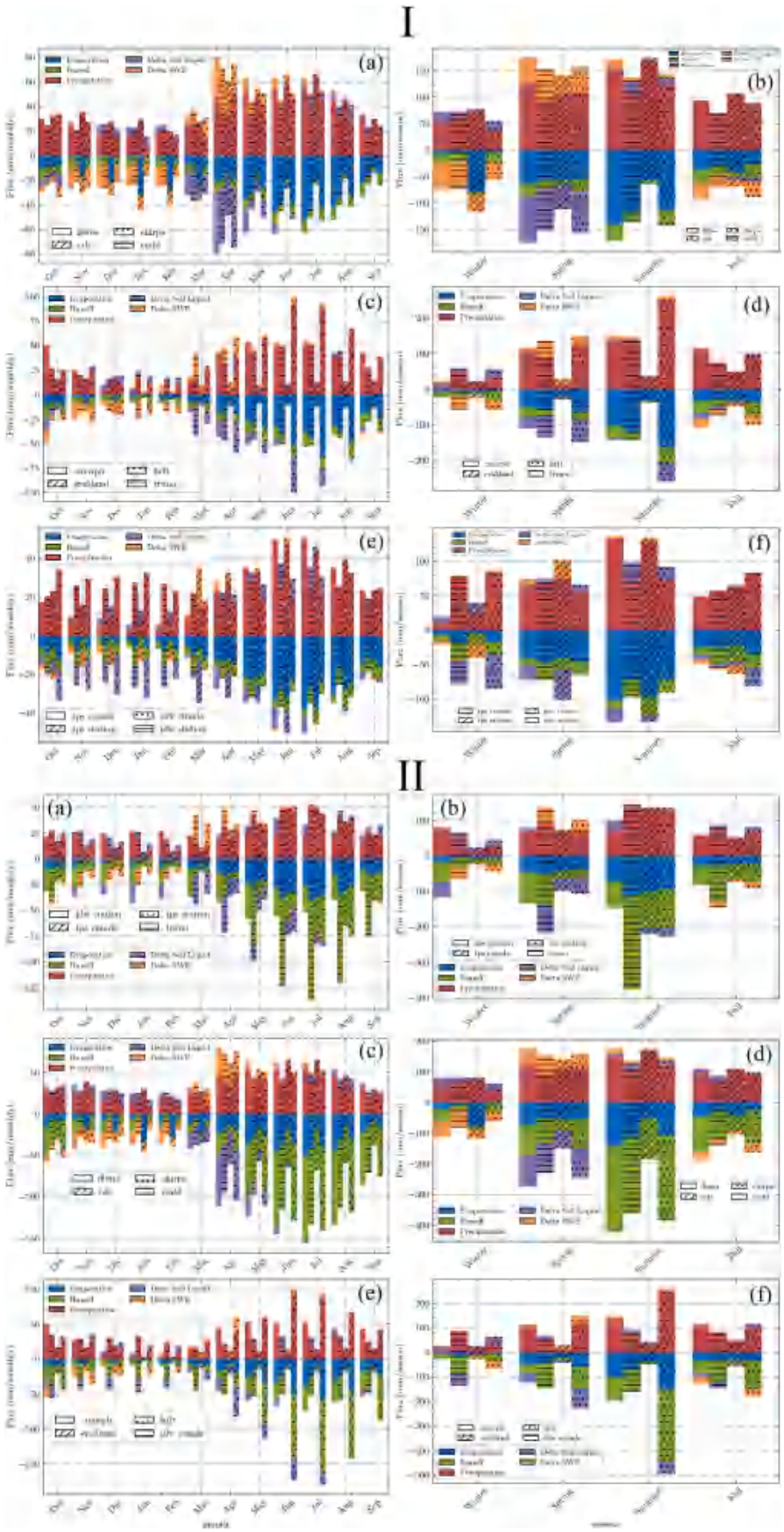

Fig. 9. Comparison of monthly and seasonal water balances in the study area (I and II are general and glacial models, respectively; precipitation in red, runoff in green, evapotranspiration in blue, change in soil moisture in purple and change in snow-water equivalent in orange; a, c and e are monthly, b, d and f are quarterly).

high water flux balances, mainly influenced by changes in precipitation and snow water equivalent, while the CMORPH and ERA5 simulations presented lower water fluxes. Water fluxes modelled by the DBMA, CHIRPS, CMFD and ERA5 data sources were significantly higher in spring and summer, dominated by evapotranspiration, whereas the CMORPH data source remained underperforming, with smaller differences in water fluxes in autumn.

In glacial mode, outputs were generally greater than inputs in spring and summer, mainly because of increased glacial meltwater. Where the DBMA simulated input water flux was 178 mm/season in spring, the output increased significantly to 278 mm/season, with the main outputs being the amount of soil moisture change, runoff and evapotranspiration in that order. In summer, the output increases even further to 424 mm/season, with runoff accounting for more than 60 % and evapotranspiration contributing significantly. In autumn, the output was reduced to 185 mm/season, and ERA5 modelled relatively low equilibrium values in all seasons, especially in winter, when the input was only 80 mm/season and the output 118 mm/season. In summary, the increased water fluxes in spring and summer mainly reflect the contribution of glacial meltwater to the hydrological regime, while the relatively low outputs in autumn and winter point to seasonal changes in evapotranspiration and precipitation.

The monthly time series of hydrological flux simulations showed that in the evapotranspiration simulations, the HRLT data exhibited relatively high and abundant high-frequency variations throughout the year, followed by the DBMA, respectively, where CMFD and CHIRPS were close to each other (Fig. 10). In contrast, the PERSIANN simulations showed more chaotic variations, with TRMM relatively high in spring and summer, relatively low in autumn and winter, and overall low in CMORPH and ERA5. Surface soil moisture simulations showed that it remained at medium-high values from October to December, declined rapidly from January to March, and then showed an increasing trend from March onwards until it plateaued. HRLT had the highest performance in October to December, followed by DBMA, CHIRPS and CMFD. From March to September, the data rose rapidly, except for HRLT, which initially grew slowly, and ultimately HRLT was the highest performer, followed by DBMA, CHIRPS, and CMFD. Snow water equivalent (SWE) simulations followed a pattern of high in autumn and winter and low in spring and summer, with DBMA, CHIPRS and CMFD peaking rapidly from October to February, DBMA decreased more slowly during March to September, PERSIANN showed higher performance, and CMORPH peaked earlier in winter. In the runoff simulation, HRLT performance was relatively high from March to April, and the rest of the data showed different high-frequency variations.

In the glacier model, the evapotranspiration simulation results showed more high-frequency variations in CMFD and CHIRPS, and HRLT and DBMA were close to each other, which showed low in autumn and winter and high in spring and summer. CMORPH fluctuated significantly, ERA5 and TRMM were closer, PERSIANN was the smallest, and there was no clear seasonal pattern. In the surface soil moisture simulation results, except for CMFD, the rest of the data showed a seasonal variation pattern like that of the common model, with CHIRPS relatively high in most seasons, followed by HRLT and DBMA. The snow water equivalent (SWE) simulations were consistent with a pattern of high autumn/winter and low spring/summer, with the DBMA simulations continuing to be low after a rapid peak in autumn/winter. The runoff simulation results showed that the data were low in March–April except for HRLT, CHIRPS was relatively high in April–September, CMFD showed spiral fluctuations, HRLT peaked at the end of July, and DBMA rose fast but peaked small. CMORPH showed no clear seasonal pattern and TRMM was highest in late July. The interpolated data showed a similar seasonal pattern, with large and high variability in TPS STATION and high frequency in TPS CMADS.

4. Discussion

4.1. Uncertainty analysis

The greatest uncertainty in hydrological modeling originates from the uncertainty in meteorologically driving data, especially precipitation inputs (Groisman and Legates, 1994; Wu et al., 2023). Traditional meteorological driven data has experienced a shift from station interpolation to the increasing number of satellite precipitation data products available today, and even the fusion of precipitation data as the driving data for hydrological models (Jahanshahi et al., 2024; Tang et al., 2021). It has been shown that as the scale increases from site to interpolated gridded datasets (scales of 10^3–10^4 m), precipitation uncertainty increases from 5-16 % to 33–65 % of the rainfall rate (McMillan et al., 2012). Remotely sensed satellite precipitation products can compensate for the shortcomings of station data, but they suffer from a shortage of adequate spatial and temporal resolution and a lack of accuracy in quantifying solid-phase precipitation and precipitation in mountainous areas (Lettenmaier, 2017). With the development of Markov chain Monte Carlo methods and advances in computational power, Bayesian inference or ensemble-based methods have become popular for quantifying uncertainty in hydrological modeling caused by uncertain model inputs, parameters or structure (Rahman et al., 2020; Tang et al., 2019).

In this study, the DREAM algorithm was used to demonstrate its remarkable ability to capture runoff modeling uncertainty, with the modelled 95% prediction intervals covering more than 70% of the measured runoff in most cases. Firstly, at the Wenquan Hydrological Station, the DBMA, CHIRPS, CMFD, TRMM, and TPS CMADS data showed good predictive ability by predicting intervals similar to the measured runoff trends during the simulation period (Fig. 11). Of these, the DBMA modelled values had wide forecast ranges up to 2011, showing greater uncertainty. In contrast, PERSIANN, CMORPH, ERA5, and HRLT performed poorly with large widths of prediction intervals at specific times but differed significantly from measured runoff trends. Secondly, at Jinghe Hydrological Station, the prediction intervals of DBMA, CHIRPS and HRLT were more consistent with the trend of the observed data. In contrast, the CDR, CMORPH, and ERA5-Land data performed relatively poorly, with wide prediction intervals and inconsistencies with measured data. Finally, at the Bole hydrological station, the prediction intervals of the IDW CMADS and TPS CMADS covered the measured runoff better, but the seasonal fluctuation trends were not in complete agreement with the observations. In addition, TPS STATION, HRLT, and TRMM data, although the prediction intervals could include the measured runoff in some years, generally suffered from mismatch of seasonal fluctuations, probably because Bole hydrological station is in the periphery of the city and has a high impact of human activities. In summary, DBMA, CHIRPS and HRLT provided more accurate modeling results to a certain extent, whereas the data of PERSIANN, CMORPH and ERA5 were more uncertain.

Although uncertainty in runoff simulation from different data sources was initially identified after algorithmic iterations, the uncertainty in hydrological modeling was not further quantified by factors such as model structure and parameters. In terms of hydrological model structure, the VIC model in combination with the glacier module can significantly improve the simulated flow rate at peak summer runoff, and previous studies have reported a similar situation (Zhao et al., 2015, 2019). However, in the face of areas that were heavily influenced by human activities (Bole hydrological station), the accuracy was generally about 30% lower than that of the outflow hydrological stations (Jinghe and Wenquan hydrological stations). The VIC model mainly simulates the changes of hydrological variables in the natural state and lacks accurate simulation of hydrological systems under the influence of human activities, so it needs to be further improved, for example, by adding a module on agricultural irrigation and a module on socio-economic water use. In addition to the model structure, the calibration of model parameters is also an important factor affecting the simulation accuracy

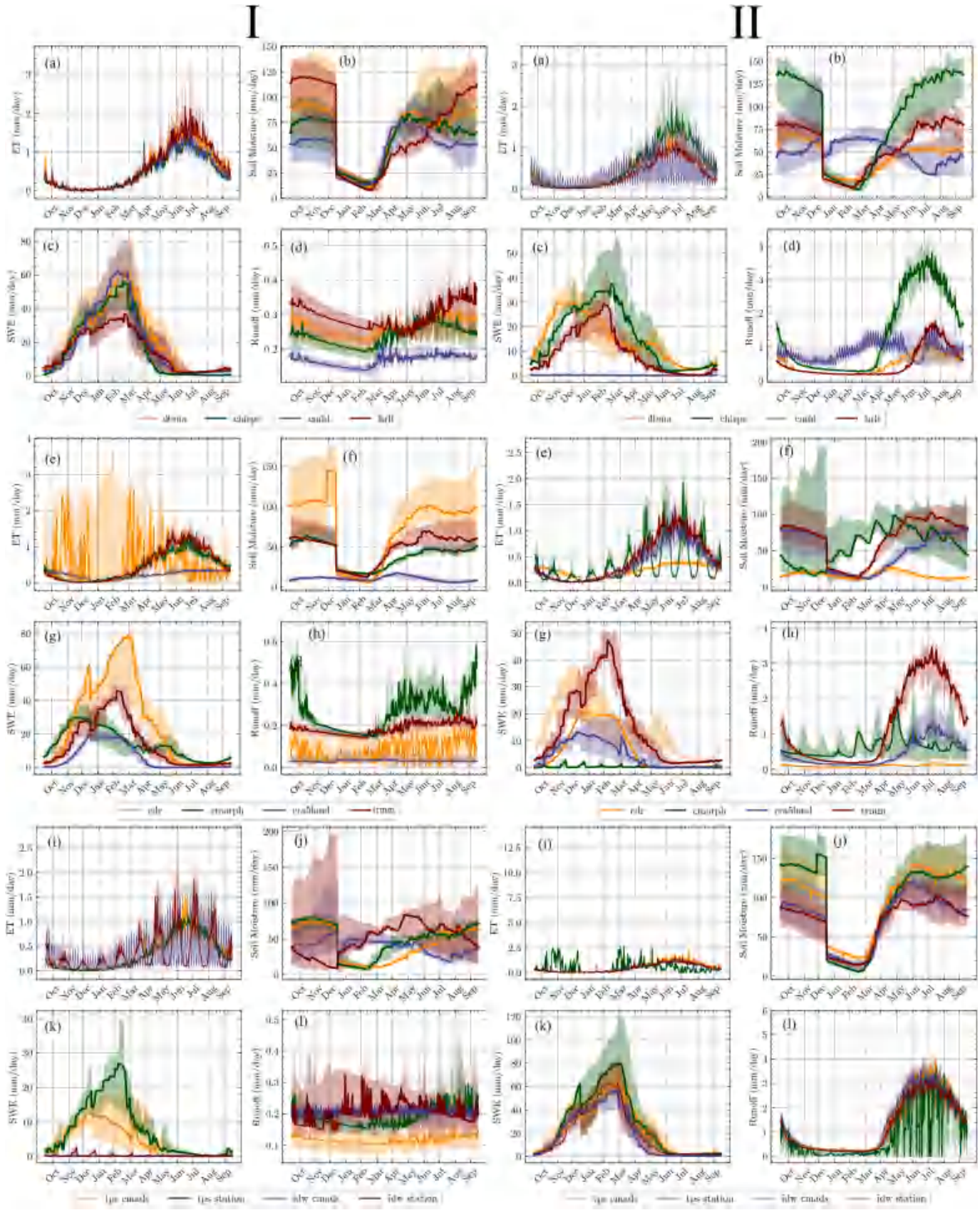

Fig. 10. Comparison of time series of modelled hydrological variables in the study area (I and II for general and glacial models, respectively).

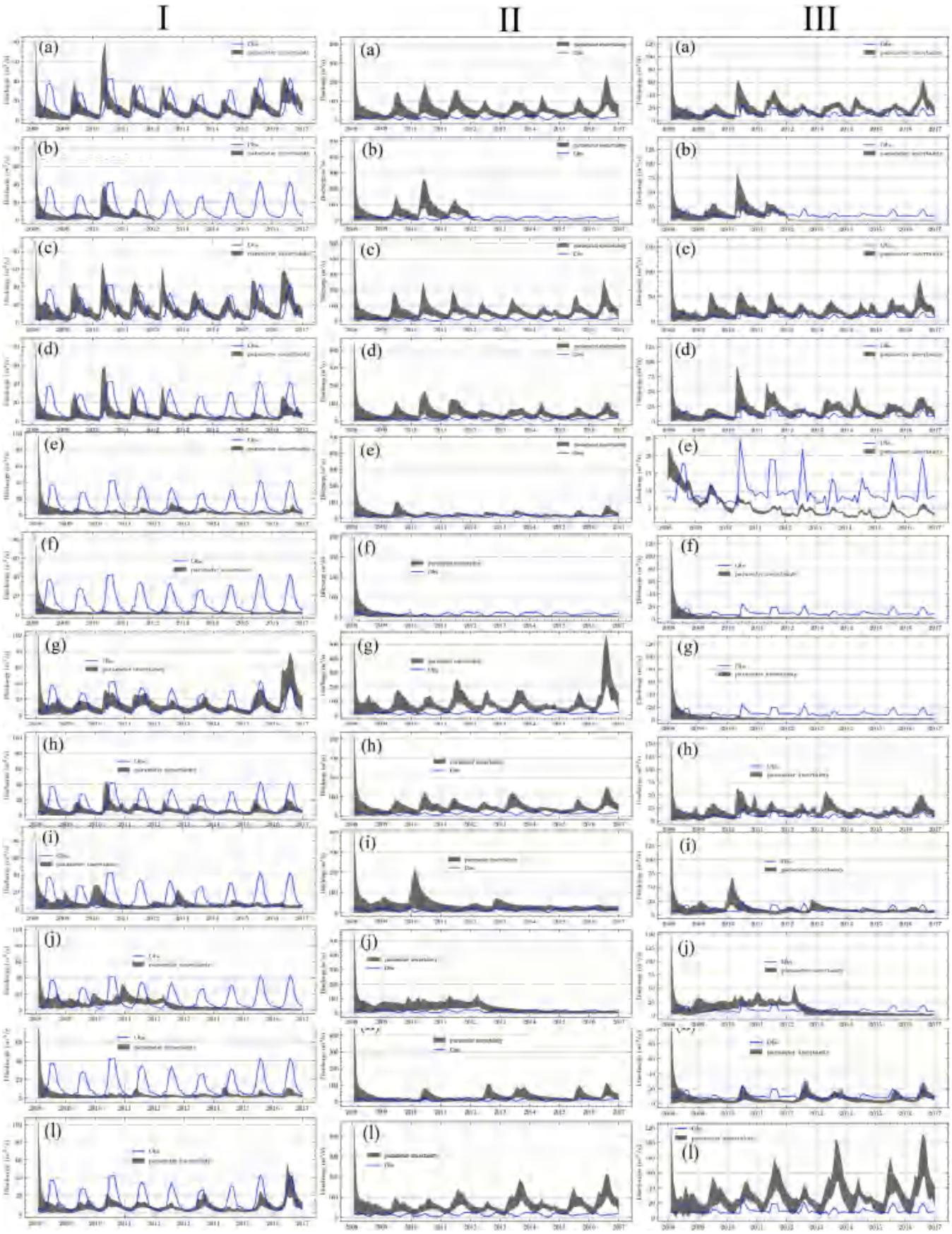

Fig. 11. Posterior uncertainty of runoff simulation. (Where I, II and III are Jinghe, Bole and Wenquan Hydrological Stations, respectively; a, b, c, d, e, f, g, h, i, j, k and l are DBMA, PERSIANN, CHIRPS, CMFD, CMORPH, ERA5, HRLT, TRMM, IDW CMADS, IDW STATION, TPS CMADS, and TPS STATION respectively.).

(Chegwidden et al., 2019; Hamman et al., 2018). The VIC model, being a continuous-time hydrological model with the capability of outputting multiple hydrological fluxes at the same time, has traditionally been calibrated and validated based on continuous-time data from runoff observations, and statistical metrics, such as NSE and R^2, have been used to assess the strengths and weaknesses of the simulation accuracy (Mendoza et al., 2015). Numerous studies have demonstrated that the VIC hydrological model works satisfactorily for runoff-based continuous hydrological models in different geoclimatic regions around the world (Dash et al., 2021; Scheidegger et al., 2021; Zheng et al., 2022). However, hydrological systems are non-linear, and using metrics such as NSE and R^2 as calibration functions may lead to biases in flow simulations, while only answering the question of differences in model performance, but not the in-depth answer to the question, 'What are the reasons behind these differences?' (Weijs et al., 2010). Therefore, there is need for further development of an advanced technique that can account for each behavior of the hydrological model (including non-linear, feedback and emergent behaviors) for more reliable analyses and comparisons.

4.2. Transfer entropy analysis

To answer the question in the previous subsection, this study used the time-dependent weighing of transfer entropy to quantify how much information was transferred between the water balance model fluxes. In this way it is possible to evaluate the model outputs of the hydrological system under different driving data simulations and to construct a network of processes, accordingly, thus inferring how the different driving data ultimately affect the model outputs.

The entropy of information transfer results showed that precipitation occupies a prominent position in information transfer, especially on evapotranspiration and surface soil moisture, highlighting its critical role in the water cycle. The high entropy of transfer between evapotranspiration and other hydrological fluxes, especially with precipitation, indicated that the evapotranspiration process is directly influenced by precipitation events. It is because of the climatic characteristics of the arid region, where below-cloud evaporation of precipitation is significant (Zhu et al., 2021). Surface soil moisture exhibited complex information transfer relationships with several hydrological fluxes, particularly a significant link with evapotranspiration. The information transfer between snow water equivalent and other fluxes was relatively weak and was mainly influenced by precipitation and evapotranspiration.

Detailed analyses of the individual driving data simulations showed that the CMFD and TPS CMADS simulations received higher information entropy in terms of snow water equivalent, reflecting the fact that they are mainly influenced by soil moisture, runoff and precipitation (Fig. 12). Relatively, IDW CMADS and TPS STATION had low information entropy. In terms of runoff, CMFD, IDW STATION, and TPS STATION had higher information entropy, mainly contributed by snow water equivalent, evapotranspiration, and precipitation, while IDW CMADS, DBMA, and ERA5 had lower information entropy. For surface soil moisture, DBMA, HRLT and IDW CMADS received higher information entropy, mainly contributed by runoff, precipitation and evapotranspiration, while CHIRPS and TPS CMADS were lower. In terms of evapotranspiration, CHIRPS and TPS CMADS received higher information entropy and were mainly influenced by precipitation, runoff and soil moisture, while CMFD, CMORPH and IDW CMADS were lower. For precipitation, IDW CMADS and TPS CMADS had higher information entropy, showing significant contributions from snow water equivalent, soil moisture, and runoff, while IDW STATION and PERSIANN had lower information entropy.

The transfer of information on runoff was not only influenced by precipitation and surface soil moisture, but also involved evapotranspiration and snow water equivalent, suggesting that runoff was influenced by multiple factors in the glacial model. Evapotranspiration was mainly influenced by soil moisture and snow water equivalent (SWE) in the glacier model, as opposed to precipitation and runoff in the general model, suggesting that evapotranspiration in the glacier model was more dependent on soil moisture and SWE. The information transfer of precipitation in the glacial model was more closely linked to evapotranspiration and snow water equivalent, while the link to surface soil moisture was weakened. Surface soil moisture was more influenced by snow water equivalent and evapotranspiration in glacial mode. Specifically for the different modelled data, PERSIANN, CHIRPS, ERA5 and IDW STATION received higher information entropy in terms of snow water equivalent, mainly contributed by precipitation, runoff and evapotranspiration together, while DBMA, TRMM and IDW CMADS were lower. In terms of runoff, DBMA and CMORPH received higher information entropy, showing significant effects of precipitation, snow water equivalent and surface soil moisture. In soil moisture, CHIRPS and TRMM showed high information entropy, mainly influenced by snow water equivalent, evapotranspiration and runoff. For evapotranspiration, ERA5, HRLT, and TPS CMADS had higher information entropy, reflecting their strong linkages with runoff, precipitation, and soil moisture. For precipitation, DBMA, CHIRPS, TRMM, and IDW STATION had higher information entropy, which was mainly influenced by soil moisture, runoff, and snow water equivalent.

5. Conclusions

In this study, six mainstream precipitation products were integrated by the DBMA algorithm, followed by hydrological modeling using different driving data including site interpolated data combined with the VIC model, parameter calibration optimization by the DREAM algorithm and data assimilation by Kalman filtering to improve the simulation accuracy. To discuss uncertainty in hydrological modeling, a process-level assessment of modelled hydrological fluxes was also carried out using information transfer entropy. The main conclusions were as follows.

(1) The DBMA fusion weights showed that CMFD (0.57) and ERA5-Land (0.38) were assigned relatively high weights; CMFD had a higher weight compared to spring and winter, whereas CHIRPS received a lower weight assignment in summer and autumn. The annual mean value of DBMA fusion precipitation was around 309.83 mm, with average precipitation of 11.32 mm, 54.71 mm, 31.86 mm and 22.11 mm in spring, summer, autumn and winter, respectively. The inter-annual temporal trends were largely suppressed by the measured values, ranging from 0 to 120 mm, with intra-annual variations ranging from 0 to 50 mm, with a significant underestimation in June. The DBMA fusion precipitation achieved good accuracy with measured values of RMSE, R^2, SD and BIAS wield 1.46, 0.75, 1.65 and 6.00, respectively.
(2) After parameter calibration optimization and data assimilation, all the driving data were simulated to a satisfactory level (NSE >0.80). CHIRPS and TRMM-driven simulations were relatively better in normal mode, with NSEs of 0.18 and 0.15, respectively; comparing with glacier mode, HRLT and CMFD simulations were relatively better, with NSEs of 0.77 and 0.73, respectively, and DBMA-fused precipitation was worse instead.
(3) The DBMA, CHIRPS and HRLT simulations showed higher runoff values in the south-eastern and western regions with mean values of 53.22 mm, 50.83 mm and 65.54 mm, respectively. In glacial mode, runoff was significantly higher in the surrounding mountains than in the rest of the region. Seasonal water balance analyses showed that in winter, the DBMA, CMFD and CDR data had the highest simulated water flux balance values, while ERA had the lowest; in spring and summer, simulated water fluxes were significantly higher for DBMA, CHIRPS, CMFD and ERA5 than for the other data sources, with smaller differences in autumn. In the glacial model, DBMA simulated water fluxes increased from 178 mm/season and 278 mm/season to 424 mm/season in spring and

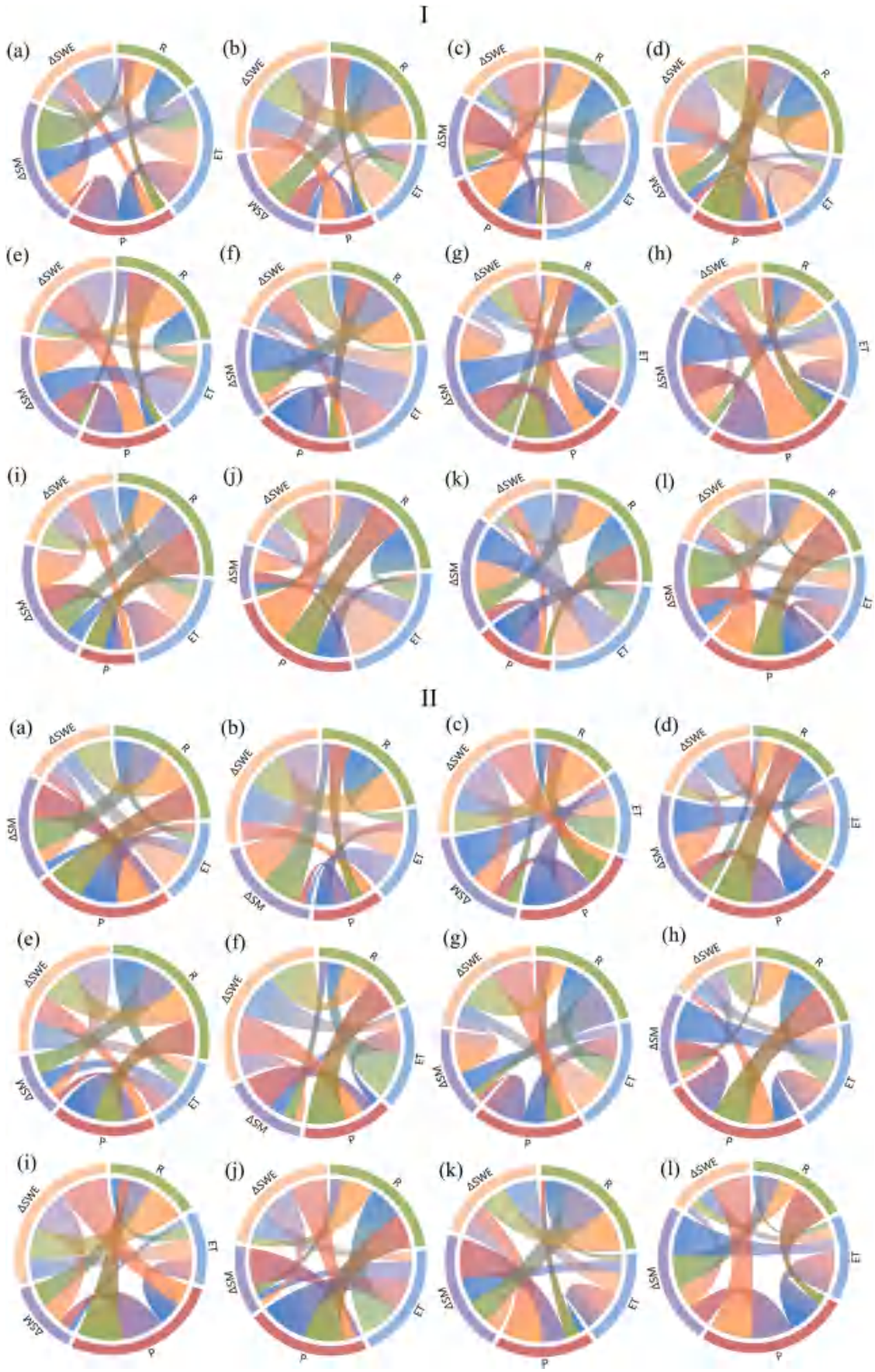

Fig. 12. Entropy chord plot of lag 1-day transfer in the study area. (a, b, c, d, e, f, g, h, i, j, k and l for DBMA, PERSIANN, CHIRPS, CMFD, CMORPH, ERA5, HRLT, IDW CMADS, IDW STATION, TPS CMADS, TPS STATION, and TRMM, respectively; and I and II for Normal Mode and Glacier Mode, respectively).

summer due to glacial meltwater effects, with runoff and evapotranspiration as the main outputs, while ERA5 simulated equilibrium values were lower in all seasons, with inputs of only 80 mm/season in winter.

(4) The DREAM algorithm showed that the DBMA, CHIRPS and HRLT data performed better by covering more than 70% of the measured runoff in the 95% prediction interval of the hydrological station simulation, while the PERSIANN, CMORPH and ERA5 data were more uncertain, and the prediction results differed significantly from the measured runoff. The information transfer entropy results showed that precipitation plays a significant role in information transfer, particularly affecting evapotranspiration and surface soil moisture; CMFD and TPS CMADS simulations had the highest information entropy for snow-water equivalent, while CHIRPS and TPS CMADS had the highest information entropy for evapotranspiration, showing significant contributions from precipitation, runoff and soil moisture. In the glacial model, runoff was influenced by multiple factors, including precipitation, surface soil moisture, evapotranspiration and snow water equivalent, with PERSIANN, CHIRPS, ERA5 and IDW STATION having the highest entropy of snow water equivalent information. DBMA and CMORPH had the highest runoff information entropy, CHIRPS and TRMM.

CRediT authorship contribution statement

Qingling Bao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Funding acquisition. **Jianli Ding:** Supervision, Funding acquisition, Conceptualization. **Jinjie Wang:** Project administration, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envsoft.2025.106376.

Data availability

Data will be made available on request.

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